

MH2100 Calculus III

Midterm I (Solutions)

AY 2018/2019, Semester 1

Quantitative Research Society @NTU

September 2018

Overview

- **Coverage:** parametric curves and vector calculus; partial derivatives, limits, continuity and differentiability; critical points and the second-derivative test.
- **Format:** three questions, 100 marks, 90 minutes (official paper: September 2018).
- **Historical permitted material:** the September 2018 paper permitted a two-sided A4 piece of paper with handwritten notes. Consult current course materials for present assessment rules.

Instructions (verbatim)

1. DO NOT TURN OVER PAPER UNTIL INSTRUCTED.
2. This midterm paper contains THREE (3) questions.
3. Answer ALL questions. The marks for each question are indicated at the beginning of each question.
4. Candidates can write anywhere on this midterm paper.
5. A two-sided A4 piece of paper with handwritten notes is permitted.

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Question 1**(50 Marks)**

Circle (only) the correct answer for each question. Each question is worth 5 marks.

- (i) Find parametric equations to represent the line segment from $(-3, 4)$ to $(12, -8)$.

- (a) $x = -3 - 15t, \quad y = 4 - 12t, \quad 0 \leq t \leq 1$
- (b) $x = -3 + 15t, \quad y = 4 - 12t, \quad 0 \leq t \leq 2$
- (c) $x = 8 - 15t, \quad y = 4 - 12t, \quad 0 \leq t \leq 2$
- (d) $x = -3 + 15t, \quad y = 4 - 12t, \quad 0 \leq t \leq 1$
- (e) $x = -3 - 15t, \quad y = 4 - 12t, \quad 0 \leq t \leq 2$

- (ii) Find the point(s) on the following curve where the tangent is horizontal.

$$x = t^3 - 3t + 2, \quad y = t^3 - 3t^2 + 2$$

- (a) $(0, 0)$
 - (b) $(0, 0)$ and $(2, -2)$
 - (c) $(2, 2)$ and $(-2, -2)$
 - (d) $(2, 2)$ and $(4, -2)$
 - (e) None of the above
- (iii) Find parametric equations for the tangent line at the point when $t = 1$ to the curve with parametric equations

$$x = 3t, \quad y = 5t^2, \quad z = 4t^3.$$

- (a) $x = 1 + 3t, \quad y = 1 + 10t, \quad z = 1 + 12t$
 - (b) $x = 3 + 3t, \quad y = 5 + 10t, \quad z = 4 + 12t$
 - (c) $x = 1 + 3t, \quad y = 1 + 5t, \quad z = 1 + 4t$
 - (d) $x = 3 + 3t, \quad y = 5 + 5t, \quad z = 4 + 4t$
 - (e) None of the above
- (iv) Find the unit tangent vector for the curve given by

$$\mathbf{r}(t) = \left\langle \frac{1}{5}t^5, \frac{1}{3}t^3, t \right\rangle.$$

- (a) $\frac{\langle t^4, t^2, 1 \rangle}{\sqrt{t^{10} + t^4}}$
 - (b) $\frac{\langle t^4, t^2, 1 \rangle}{\sqrt{4t^8 + 4t^4 + 1}}$
 - (c) $\frac{\langle t^4, t^2, 1 \rangle}{\sqrt{t^8 + t^4 + 1}}$
 - (d) $\frac{\langle t^4, t^2, 1 \rangle}{\sqrt{t^8 + t^4}}$
 - (e) None of the above
- (v) Find the directional derivative of the function $f(x, y, z) = xyz$ in the direction of the vector $\langle 5, -3, 2 \rangle$.

- (a) $\frac{1}{\sqrt{38}}(5yz - 3xz + 2xy)$
- (b) $\frac{1}{\sqrt{38}}(5xy - 3xz + 2yz)$
- (c) $5yz - 3xz + 2xy$
- (d) $5xy - 3xz + 2yz$
- (e) None of the above
- (vi) Find the gradient vector of the function $f(x, y, z) = z^7 e^{2x\sqrt{y}}$.
- (a) $e^{x^2\sqrt{y}} \left\langle 7xz\sqrt{y}, \frac{x^2 z}{2\sqrt{y}}, 2 \right\rangle$
- (b) $ze^{2x\sqrt{y}} \left\langle z\sqrt{y}, \frac{xz}{2\sqrt{y}}, 7z^5 \right\rangle$
- (c) $e^{x^2\sqrt{y}} \left\langle 2xz\sqrt{y}, \frac{x^2 z}{7\sqrt{y}}, 2 \right\rangle$
- (d) $ze^{2x\sqrt{y}} \left\langle 2z^6\sqrt{y}, \frac{xz^6}{\sqrt{y}}, 7z^5 \right\rangle$
- (vii) Use the Chain Rule to find $\frac{dw}{dt}$ for
- $$w = 3x^6 y^3 z, \quad x = 6t, \quad y = \cos 2t, \quad z = t \sin t.$$
- (a) $3x^5 y^2 (36yz - 6xz \sin 2t + xy(\sin t + t \cos t))$
- (b) $3x^6 y^3 (36yz - 6xz \sin 2t + xy(\sin t + t \cos t))$
- (c) $648x^5 y^2 \sin 2t (\sin t + t \cos t)$
- (d) $648x^6 y^3 \sin 2t (\sin t + t \cos t)$
- (e) None of the above
- (viii) Find an equation for the tangent plane to the surface given by the equation $xy + yz + xz = 38$ at the point $(4, 2, 5)$.
- (a) $7x + 9y + 6z = 70$
- (b) $7x + 9y + 6z = 74$
- (c) $7x + 9y + 6z = 76$
- (d) $7x + 9y + 6z = 78$
- (e) None of the above
- (ix) Find an equation for the tangent plane to the surface $z = 4x^3 y^2 + 2y$ at the point $(1, -2, 12)$.
- (a) $48x + 14y - z = 60$
- (b) $48x - 14y - z = 64$
- (c) $48x - 14y - z = 60$
- (d) $48x + 14y - z = 64$

- (e) None of the above
- (x) Find and classify the critical points of the function $f(x, y) = e^{-2x} \sin 4y$ for $x \geq 0$ and $0 \leq y \leq \pi/2$.
- (a) Local maximum $f(0, \pi/8) = 1$
- (b) Saddle point $f(0, 0) = 0$
- (c) Local minimum $f(0, \pi/4) = 0$
- (d) None of the above

Solution

Method 1: Direct computation

- (i) $\mathbf{p} + t(\mathbf{q} - \mathbf{p})$ with $\mathbf{p} = (-3, 4)$, $\mathbf{q} - \mathbf{p} = (15, -12)$, $0 \leq t \leq 1$:

$$(d) \quad x = -3 + 15t, \quad y = 4 - 12t, \quad 0 \leq t \leq 1.$$

- (ii) $y' = 3t(t - 2) = 0 \iff t = 0, 2$, and $x' = 3(t^2 - 1) \neq 0$ there; $t = 0, 2 \mapsto (2, 2), (4, -2)$:

$$(d) \quad (2, 2) \text{ and } (4, -2).$$

- (iii) $\mathbf{r}(1) = (3, 5, 4)$, $\mathbf{r}'(1) = (3, 10, 12)$: (b) $x = 3 + 3t$, $y = 5 + 10t$, $z = 4 + 12t$.

- (iv) $\mathbf{r}' = \langle t^4, t^2, 1 \rangle$, $\|\mathbf{r}'\| = \sqrt{t^8 + t^4 + 1}$:

$$(c) \quad \mathbf{T}(t) = \frac{\langle t^4, t^2, 1 \rangle}{\sqrt{t^8 + t^4 + 1}}.$$

- (v) $\nabla f = \langle yz, xz, xy \rangle$, $\mathbf{u} = \langle 5, -3, 2 \rangle / \sqrt{38}$:

$$(a) \quad D_{\mathbf{u}}f = \frac{5yz - 3xz + 2xy}{\sqrt{38}}.$$

- (vi) For $y > 0$:

$$\nabla f = e^{2x\sqrt{y}} \left\langle 2z^7\sqrt{y}, \frac{xz^7}{\sqrt{y}}, 7z^6 \right\rangle = ze^{2x\sqrt{y}} \left\langle 2z^6\sqrt{y}, \frac{xz^6}{\sqrt{y}}, 7z^5 \right\rangle \implies (d).$$

- (vii) Chain Rule, $w_x = 18x^5y^3z$, $w_y = 9x^6y^2z$, $w_z = 3x^6y^3$:

$$\begin{aligned} \frac{dw}{dt} &= 6w_x - 2\sin 2t w_y + (\sin t + t \cos t) w_z \\ &= 3x^5y^2(36yz - 6xz \sin 2t + xy(\sin t + t \cos t)) \implies (a). \end{aligned}$$

- (viii) $\nabla(xy + yz + xz)(4, 2, 5) = \langle 7, 9, 6 \rangle$:

$$7(x - 4) + 9(y - 2) + 6(z - 5) = 0 \implies (c) \quad 7x + 9y + 6z = 76.$$

- (ix) $f_x = 12x^2y^2 = 48$, $f_y = 8x^3y + 2 = -14$ at $(1, -2)$:

$$z - 12 = 48(x - 1) - 14(y + 2) \implies (b) \quad 48x - 14y - z = 64.$$

- (x) Literal: critical point $:=$ interior point with $\nabla f = 0$, but $f_x = -2e^{-2x} \sin 4y$, $f_y = 4e^{-2x} \cos 4y$ never both vanish $\implies$ none (d).

Intended: $(0, \pi/8)$ lies on the boundary $x = 0$; extrema on the domain, boundary included. $f(0, \pi/8) = 1$, $f \leq e^{-2x} \leq 1$ ($x \geq 0$) $\implies$ local max.

intended: (a) Local maximum $f(0, \pi/8) = 1$;	literal: (d).
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Unofficial QRS marking scheme (50 marks)

- (i) **(5 marks)** 2: correct displacement; 2: correct parameter interval; 1: selects (d).
- (ii) **(5 marks)** 2: $y' = 0$; 2: checks $x' \neq 0$ and finds both points; 1: selects (d).
- (iii) **(5 marks)** 2: point; 2: tangent vector; 1: selects (b).
- (iv) **(5 marks)** 2: derivative; 2: norm; 1: selects (c).
- (v) **(5 marks)** 2: gradient; 2: normalized direction; 1: selects (a).
- (vi) **(5 marks)** 4: all gradient components; 1: selects (d).
- (vii) **(5 marks)** 4: correct chain-rule terms and simplification; 1: selects (a).
- (viii) **(5 marks)** 2: normal; 2: plane through the point; 1: selects (c).
- (ix) **(5 marks)** 2: partial derivatives; 2: plane; 1: selects (b).
- (x) **(5 marks)** 2: identifies the boundary-definition issue; 2: constrained local-maximum argument; 1: states intended (a) and literal (d).

Question 2**(25 Marks)**

- (i) Evaluate each of the following limits or show that the limit does not exist. Justify your answers.

(a) $\lim_{(x,y) \rightarrow (6,6)} \frac{5e^{x-y}}{\cos(\pi(x-y))}$

(b) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^{21}y^{21}}{x^{42} + y^{84}}$

- (ii) Define

$$f(x, y) = \begin{cases} (x^2 + y^2) \sin\left(\frac{1}{x^2 + y^2}\right); & \text{if } x^2 + y^2 \neq 0, \\ 0; & \text{if } x^2 + y^2 = 0. \end{cases}$$

- (a) Is f continuous at $(0, 0)$? Justify your answer.
 (b) Find $f_x(0, 0)$ and $f_y(0, 0)$.
 (c) Is f differentiable at $(0, 0)$? Justify your answer.

Solution**Method 1: Direct limit and derivative definitions**

- (i) (a) Quotient continuous at $(6, 6)$ (numerator $5e^0 = 5$, denominator $\cos 0 = 1 \neq 0$):

$$\lim_{(x,y) \rightarrow (6,6)} \frac{5e^{x-y}}{\cos(\pi(x-y))} = 5.$$

- (b) Path $y = 0$: quotient = 0. Path $y = x \neq 0$: $\frac{x^{42}}{x^{42} + x^{84}} = \frac{1}{1 + x^{42}} \rightarrow 1 \neq 0$.

the limit does not exist.

- (ii) (a) $|f| \leq x^2 + y^2 \rightarrow 0 = f(0, 0)$ since $|\sin u| \leq 1$ (Squeeze): f is continuous at $(0, 0)$.

(b) $f_x(0, 0) = \lim_{h \rightarrow 0} \frac{h^2 \sin(1/h^2)}{h} = \lim_{h \rightarrow 0} h \sin(1/h^2) = 0;$

$$f_y(0, 0) = \lim_{k \rightarrow 0} \frac{k^2 \sin(1/k^2)}{k} = \lim_{k \rightarrow 0} k \sin(1/k^2) = 0 :$$

$$f_x(0, 0) = f_y(0, 0) = 0.$$

- (c) $L = (f_x, f_y)(0, 0) = 0$ is the only candidate; with $r = \sqrt{h^2 + k^2}$:

$$\frac{|f(h, k) - f(0, 0) - L(h, k)|}{r} = r \left| \sin\left(\frac{1}{r^2}\right) \right| \leq r \rightarrow 0$$

$$f \text{ is differentiable at } (0, 0), \quad Df(0, 0) = 0.$$

Unofficial QRS marking scheme (25 marks)**(i) (10 marks)**

- (a) **(5 marks)** 2: recognizes continuity of the quotient; 2: evaluates numerator and denominator; 1: final value 5.
- (b) **(5 marks)** 2: path $y = 0$; 2: path $y = x$; 1: concludes nonexistence.

(ii) (15 marks)

- (a) **(5 marks)** 3: valid squeeze bound; 2: continuity conclusion.
- (b) **(5 marks)** 2 each: correct difference quotients; 1: both values equal zero.
- (c) **(5 marks)** 2: identifies $L = 0$; 2: normalizes the remainder; 1: differentiability conclusion.

Question 3**(25 Marks)**

Let

$$f(x, y) = y^3 - x^2 - 7y^2 + 2xy + 9y.$$

Find and classify all critical points of f . (For each critical point, determine whether or not it is a local max/min or a saddle point.)

Solution**Method 1: Direct derivative and second-derivative test**

$$\begin{aligned} f_x = -2x + 2y = 0 &\implies x = y \\ f_y = 3y^2 - 14y + 2x + 9 = 3y^2 - 12y + 9 = 3(y-1)(y-3) = 0 &\quad (x = y) \end{aligned}$$

Critical points: $(1, 1)$, $(3, 3)$.

$$f_{xx} = -2, \quad f_{xy} = 2, \quad f_{yy} = 6y - 14 \implies D = f_{xx}f_{yy} - f_{xy}^2 = 24 - 12y$$

$(1, 1)$: $D = 12 > 0$, $f_{xx} < 0 \implies$ local maximum, $f = 4$. $(3, 3)$: $D = -12 < 0 \implies$ saddle, $f = 0$.

$(1, 1)$ is a local maximum; $(3, 3)$ is a saddle point.

Method 2: Completing the square

$u := x - y$ (invertible linear change), $f = y(y-3)^2 - u^2$:

$$\nabla f = 0 \iff u = 0, \quad \frac{d}{dy}[y(y-3)^2] = 3(y-1)(y-3) = 0 \implies (1, 1), (3, 3).$$

$$\begin{aligned} f &= y(y-3)^2 - (x-y)^2 \\ (1, 1) : f &= 4 - u^2 - 3v^2 + v^3 \leq 4 - u^2 - 2v^2 < 4 \quad (v := y-1, |v| < 1, (u, v) \neq 0) \\ (3, 3) : f &= 3v^2 + v^3 - u^2 \quad (v := y-3) \end{aligned}$$

$(1, 1)$: $f < 4 = f(1, 1)$ nearby $\implies$ strict local max. $(3, 3)$: $f(3, 3) = 0$, $f > 0$ on $u = 0$, $v \neq 0$ small, $f < 0$ on $v = 0$, $u \neq 0 \implies$ saddle.

Unofficial QRS marking scheme (25 marks)

The breakdown follows Method 1; an equivalent rigorous structural proof, such as Method 2, receives the corresponding full credit.

- **8 marks:** computes f_x, f_y and correctly solves the simultaneous equations.
- **5 marks:** identifies both critical points $(1, 1)$ and $(3, 3)$.
- **6 marks:** computes the Hessian entries and discriminant correctly.
- **3 marks:** applies the test to classify $(1, 1)$ as a local maximum.
- **3 marks:** applies the test to classify $(3, 3)$ as a saddle point.